PassFrame tuition

Statistics & Averages

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: Find the mean of 5, 9, 7, 11, 3.
Method & answer: 5+9+7+11+3=35, 35÷5=7. Answer: 7.

Standard: A table shows books read by 6 pupils: 4, 8, 6, 8, 5, 9. Find the mode and the median.
Method & answer: Mode = 8 (appears twice). Median: order 4,5,6,8,8,9 → middle two are 6 and 8, average = 7.

Advanced: A pie chart shows a survey of 60 pupils' favourite colour: Blue=150°, Red=120°, Green=90°. How many pupils chose green?
Method & answer: Green = 90/360 × 60 = 15 pupils.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the mean of 6, 10, 8, 12, 4.

2. Find the mean of 15, 19, 17.

3. Find the median of 8, 14, 5, 11, 9.

4. Find the median of 7, 15, 3, 19, 11.

5. Find the mode of 6, 9, 9, 12, 4, 9.

6. Find the mode of 11, 11, 15, 15, 15, 7.

7. Find the range of 12, 26, 18, 34, 20.

8. Find the range of 55, 40, 70, 35.

9. Which of these is the mode of 7, 7, 10, 13, 13, 13? A) 13 B) 7 C) 10 D) 11

10. A table shows pets owned by a class: Dogs=11, Cats=7, Fish=5, Birds=6. How many pets are there in total?

11. Ice-creams sold over 4 days: Mon=22, Tue=31, Wed=17, Thu=26. What is the mean number sold per day?

12. Select ALL that apply. A frequency table shows: 1 pet = 8 pupils, 2 pets = 11 pupils, 3 pets = 6 pupils. Which statements are true? A) 25 pupils were surveyed in total B) The mode number of pets owned is 2 C) The mode number of pets owned is 1 D) More pupils own 2 pets than own 3 pets

13. Five friends scored 22, 28, 31, 25, 24 in a test. What is their mean score? A) 26 B) 25 C) 27 D) 28

14. Find the mean of 8, 8, 8, 8, 13.

15. Find the range of 14, 29, 21, 6, 17.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the median of 9, 14, 6, 11.

6. A bar chart shows sales at 4 shops: A=21, B=29, C=14, D=24. What is the mean sale per shop?

11. Which of these is the range of 41, 19, 33, 52, 27? A) 33 B) 52 C) 19 D) 30

2. Find the mean of 19, 23, 15, 27.

3. A pictogram shows books read: Amy=9, Ben=13, Cara=6, Dan=16. How many books were read altogether?

4. Which of these is the median of 8, 17, 12, 25, 5? A) 12 B) 17 C) 8 D) 25

5. Find the range of 24, 52, 37, 15, 44.

7. Find the mode of 4, 7, 7, 9, 11, 7.

8. A table shows pets: Dogs=17, Cats=13, Fish=9, Birds=6. If 6 cats leave, how many pets are there altogether now?

9. Select ALL that apply. A class recorded favourite subjects: Maths=15, Art=10, Science=7. Which statements are true? A) 32 pupils were asked in total B) Maths is the mode (most popular) C) Science is the mode (most popular) D) More pupils chose Art than Maths

10. Find the mean of 11, 17, 13, 19.

12. Find the median of 9, 4, 13, 7, 17, 10.

13. A pie chart shows a survey of 40 pupils: half prefer football. How many pupils prefer football?

14. Which of these is the mean of 18, 22, 26, 30? A) 24 B) 26 C) 22 D) 28

15. Find the range of 70, 95, 82, 78.

Standard / Hard — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. Find the mean of 26, 22, 30, 18, 34.

2. Find the median of 19, 46, 12, 33, 37.

3. A pie chart shows a survey of 48 pupils: a third prefer reading, a quarter prefer sport, and the rest prefer art. How many prefer art?

4. A survey of 5 friends' weights (kg): 38, 42, 35, 40, 44. What is the range? A) 9 B) 6 C) 7 D) 44

5. Find the range of 68, 33, 62, 45, 41.

6. Six pupils scored 16, 19, 13, 22, 19, 25 in a test. Find the mode and the mean.

7. Select ALL that apply. A table shows a shop's sales over 5 days: Mon=44, Tue=59, Wed=34, Thu=69, Fri=64. Which statements are true? A) The mean daily sale is 54 B) The range of sales is 35 C) Thursday had the highest sales D) Wednesday had the lowest sales

8. Find the median of 23, 11, 29, 17, 35, 8.

9. A pie chart (360°) shows a class's pets: dogs get a fifth of the circle. What angle, in degrees, represents dogs?

10. Which of these is the mean of 24, 39, 29, 44? A) 34 B) 36 C) 32 D) 38

11. Find the mode of 48, 41, 48, 56, 48, 33.

12. A café sold 24 coffees in the morning and a third of that number in the afternoon. What was the mean number sold over the two sessions?

13. Find the range of 94, 49, 76, 62, 67.

14. Five pupils' test scores are 66, 78, 62, 85, 74. Find the median score.

15. A pie chart shows three-quarters of a class of 32 pupils prefer maths. How many pupils prefer maths? A) 24 B) 16 C) 8 D) 32